AstraFlame: Machine-Learning Design and Results

Team Agnishikha · NASA Space Apps Challenge 2026 · *Flame in Freefall* · Prachya Biswas, Data Science Lead · November 7, 2026
Code, data and the full results notebook: github.com/prachyabiswas/NASA-Space-Apps-2026 (`notebooks/AstraFlame_Full_ML_Pipeline.ipynb`)

1. Architecture

```
NASA PSI files (verbatim extracts)
   -> reproducible ETL (byte-for-byte rebuild, 890 value checks against NASA)
   -> data contract (130 checks: schema, keys, bounds, consistency, lineage)
   -> unified table (850 microgravity records + 52 Earth-gravity reference rows)
   -> task-specific preprocessing (physics features, logged exclusions, ISS-test groups)
   -> five models, chosen from the objective and scored by nested grouped cross-validation
   -> predict(): prediction + 90% range + out-of-range warning + exact Shapley values + NASA source
   -> model API (api/server.py, JSON)  ->  dashboard (NASA Models tab, live or precomputed)
```

2. NASA data used

Investigation	Record	Rows	Use
FLEX (droplet combustion, ISS)	PSI-69	274	extinction model (206 rows), burning-rate model (256 rows)
BASS-II (flame spread, ISS)	PSI-25 + Link et al. 2018	181	flame-spread model (52 rows); gas response kept as evidence
SPICE (smoke point, ISS)	PSI-107	55	smoke-point model
SAME / SAME-R (smoke aerosol, ISS)	PSI-101, PSI-102	154	particle-size model (125 rows)
BASS (burning of solids, ISS)	PSI-26	121	evidence; not modelled (descriptive outcomes)
Saffire I–III (large fires, Cygnus)	PSI-98 to PSI-100	13	evidence and gap map; too few rows to model
FLEX-2	PSI-68	raw only	not used: no measured outcome
WSTF ground tests (Earth gravity)	Hirsch et al. 2011, NTRS 20110008636	52 (22 materials)	material oxygen limits for the gauges; checked value-by-value against the paper; never used to train microgravity models

All NASA data is public domain (CC0-1.0). Units are harmonised (kPa, % O_2, mm, cm/s), every row keeps its NASA source, and the unified table has a unique key per record. One recording error found by the data contract (a heated SAME sample logged at 0 °C) was corrected and documented.

3. The five prediction tasks

Task	Target (dependent)	Inputs (independent)	Rows / ISS groups
Flame spread	spread rate over a PMMA rod (mm/s)	oxygen, opposed airflow, rod diameter	52 / 19 tests
Flame extinction	flame went out before burnout (yes/no)	fuel, pressure, oxygen, CO_2, helium, droplet size	206 / 65 series
Burning rate	burning-rate constant K (mm^2/s)	same as extinction	256 / 78 series
Smoke point	smoke-point flame length (mm)	fuel, burner diameter, coflow velocity	55 / 11 days
Smoke particle size	count-mean diameter (µm)	material, sample temperature, airflow, aging time, campaign	125 / 78 runs

4. Preprocessing

- **Physics space:** logarithms (fire follows power laws), oxygen partial pressure (oxygen that actually reaches the flame), 1/diameter (thin rods burn faster).
- **Undefined outcomes removed, with the reason logged:** 61 FLEX droplets that broke up (neither extinguished nor burned out), 7 with missing size or pressure; 29 SAME readings with failed instrument reductions or unknown inputs.
- **Groups kept:** every row carries the id of its ISS test, series or run.
- **Noise prior:** run-to-run noise measured from near-replicate tests starts the Gaussian process.
- **Guards:** a target-leak check refuses any feature that copies the target, and a leakage guard refuses any split that puts one ISS test on both sides.

5. Validation design

- **Test:** outer folds hold out whole ISS tests (leave-one-group-out for few groups, otherwise grouped 5-fold × 3 repeats).
- **Validation:** inner grouped folds inside each training set choose the model family, hyper-parameters, the GP correction and the decision threshold.
- **Why not a random split:** rows from one ISS test are near-copies; on BASS-II a random split overstates R^2 (0.588 vs 0.560 grouped).
- **Noise ceiling:** from near-replicate runs, the best R^2 any model could reach: flame spread 0.68, burning rate 0.72, smoke point 0.99, particle size 0.84.
- **Candidates:** the physics-informed GP, the physics law alone, and 7–8 tuned classical families (ridge/logistic, with interactions, k-nearest neighbours, SVM/SVR, Gaussian process, random forest, extra trees, gradient boosting), plus a top-3 ensemble.

Selection rule (set before scoring): outputs that feed the safe-cabin map need physics-consistent answers with a calibrated range, so they use the physics-informed GP; the smoke outputs are read inside the tested range, so the most accurate model is used.

6. Physics-informed Gaussian process

y (physics space) = physics law m(x) + GP correction f(x) + ISS-test offset + noise. The law is a ridge regression on physics terms; the correction is an RBF Gaussian process, kept only when grouped inner validation (negative log predictive density) shows it helps; unseen tests receive the between-test variance; the 90% range is calibrated on inner folds. Far from NASA data the correction fades and the model returns to the physics law. Gaussian processes are chosen because decisions need well-calibrated uncertainty and the data are small (52–256 rows per task), where they remain powerful while neural networks tend to be overconfident.

7. Results (held-out ISS tests)

Task	Final model	Score (95% CI)	Share of ceiling	90% coverage	Physics violations
Burning rate	Physics-informed GP (with correction)	R^2 0.71 (0.56–0.80), MAE 0.047 mm^2/s	98%	89.7%	0%
Smoke point	Ensemble: GP + SVR + ridge, physics-form fallback	R^2 0.89 (0.81–0.92), MAE 4.4 mm	89%	87%	0%
Flame spread	Physics-informed GP (correction not needed)	R^2 0.57 (0.25–0.73), MAE 0.0096 mm/s	84%	86.5%	0%
Extinction	Physics-informed GP classifier	ROC-AUC 0.87 (0.80–0.94), Brier 0.082 (baseline 0.114), balanced accuracy 0.78	not applicable (yes/no outcome)	not applicable (gives a probability)	2.4%
Particle size	Ensemble: SVR + extra trees + random forest	R^2 0.43 (0.29–0.61), MAE 0.063 µm	51%	89%	43% (v2 ensemble)

The noise ceiling and the 90% range apply to the four numeric outputs; extinction is a yes/no outcome, so it is scored by ROC-AUC, Brier score and balanced accuracy instead.

The physics-informed models beat every classical family on extinction (AUC 0.87 vs at most 0.83) and come within 0.01 R^2 of the best on burning rate; on flame spread a tuned ridge model scores slightly higher (0.62, overlapping CI), and the GP is kept for physics consistency and its range.

Extinction classifier: decision threshold

87% of the 206 droplet tests ended in extinction, so accuracy alone is misleading (always answering "goes out" scores 0.87). The yes/no threshold is chosen by balanced accuracy on grouped out-of-fold predictions; the production threshold is **0.67**. Scored with the threshold chosen inside each training set: accuracy 0.82, balanced accuracy 0.78, macro F1 0.70, F1 of the rare class ("burns to completion") 0.51, and 72% of those rare tests detected (35% at a 0.5 threshold).

8. Improvements tested

- **Extinction threshold:** adopted (above).
- **Particle-size features** (temperature relative to each material, material × temperature, fresh/aged flag) with four model types: no improvement (best 0.38 vs 0.43); current model kept.
- **GP kernel** (RBF vs Matérn 5/2 and 3/2): differences of 0.002–0.006 R^2 where the PI-GP is the final model; RBF kept.

9. Error analysis

- **Extinction:** wrong answers are rare below 21% O_2 but common at 21–25% and above (31–60%), the conditions closest to habitat air, where NASA ran few tests.
- **Particle size:** largest errors for Teflon (median 27%); burning rate: heptane (10%) vs methanol (6%); flame spread: thinner rods (15%) vs 12.7 mm rods (8%).

10. Uncertainty, domain checks and explanations

- **90% range** on every regression answer (the GP's own calibrated interval, or an interval learned from held-out residuals for the ensembles).
- **Domain check:** each input is compared with the ranges NASA tested (overall and per fuel, rod size or material) and with the distance to the nearest NASA run; outside the domain the answer carries a warning, and the smoke-point model switches to its physics form.

- **Exact Shapley values** over input groups (one-hot columns count as one input); they add up exactly to the prediction.

11. Where NASA should test next

The models recommend the next ISS experiments where their uncertainty is largest and no NASA test exists: a 3 mm methanol droplet at **32% O_2, 144 kPa** (predicted chance of survival 50%) and flame spread over a 6.4 mm rod at **21% O_2, 7.5 cm/s** (uncertainty ±37%). This is a simple rule (uncertainty × distance from existing tests), not a full Bayesian experimental design.

12. API and output format

`python api/server.py` serves `GET /health`, `GET /models`, `POST /predict/{model}` and `POST /sweep/{model}`. Inputs are validated (unknown fields, untested categories, impossible values and oversized requests are refused); no personal data is collected. The dashboard calls the API, or falls back to precomputed answers from the same models on a static host.

```
POST /predict/flex_burning_rate   {"fuel": "Heptane", "o2_pct": 21}
{ "prediction": 0.5597, "unit": "mm^2/s", "interval_90": [0.4672, 0.6522], "in_domain": true,
"warnings": [],
  "model_used": "Physics-informed Gaussian process (d2-law terms + GP correction + series offset)",
  "contributions": {"oxygen": 0.036, "CO2 diluent": 0.025, "helium diluent": -0.024, "fuel": 0.020,
...},
  "source": "NASA PSI-69 FLEX experimental table (CC0-1.0), DOI 10.60555/mbq8-0451" }
```

13. Reproducibility

- `python run_pipeline.py`: ETL check → data contract → training → evaluation → final models → figures → dashboard export → tests, with a manifest of versions and hashes.
- 18 automated tests (data contract, reproducible ETL, leakage guards, artifact integrity, prediction contract, physics behaviour, API); GitHub Actions also type-checks and builds the dashboard.
- Pinned environment (`requirements.txt`, scikit-learn 1.9.1).

14. Limitations and next steps

> The material oxygen limits come from Earth-gravity tests at 70–101 kPa, so "20 of 22 materials keep burning at 34% O_2" is a first estimate for 8.2 psia. Data are small (52–256 rows per task) and confidence intervals are wide. Particle size is the weakest task. This is decision support, not certification.

1. Run the recommended ISS tests and retrain.
2. Measure material oxygen limits directly at 8.2 psia, 34% O_2 and in reduced gravity.
3. Add lunar-gravity fire data as NASA releases it.
4. Expert review with fire-safety engineers; release the dataset, models and API as open source.

15. Sources

- NASA Physical Sciences Informatics: PSI-69 FLEX (DOI 10.60555/mbq8-0451), PSI-25 BASS-II, PSI-26 BASS, PSI-107 SPICE (DOI 10.60555/9kbh-e962), PSI-101/102 SAME, PSI-98–100 Saffire; psi.nasa.gov
- Link et al. 2018, *Scientific Reports* 8:120 (BASS-II flame spread, Table S1)
- Hirsch et al. 2011, Selected Parametric Effects on Materials Flammability Limits, NTRS 20110008636
- Harper et al. 2016, Oxygen Partial Pressure and Oxygen Concentration Flammability, NTRS 20160004937
- Norcross et al. 2015, Evidence Report: Risk of Hypobaric Hypoxia from the Exploration Atmosphere, NTRS 20150021491
- NESC 2026, Impacts of Reduced Pressure Atmospheres on ECLSS Architectures, NTRS 20260003925